DISCRETE STRUCTURES

CMSC250

Chapter 20: Fundamental Counting Principles

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SUBFundamental Counting Principles



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1 Why counting is more than listing

Many problems ask for the number of possible schedules, passwords, teams, or arrangements. Listing all outcomes is usually impractical. The first question to ask is *what exactly counts as one outcome?* Next decide whether you are making a sequence of choices, choosing between disjoint alternatives, or selecting objects without regard to order.

Key idea. Before using a formula, state (1) what is being counted, (2) whether order matters, (3) whether repetition is permitted, and (4) whether cases overlap.

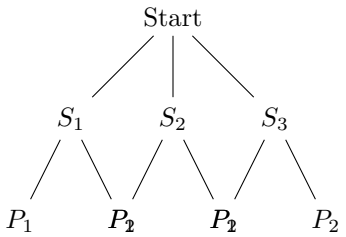
2 The multiplication principle

If a procedure has a choices for its first step and, for *each* first-step choice, b choices for its second, it has ab possible outcomes. For k successive steps with $n_1, \ldots, n_k$ possibilities, the total is $n_1 n_2 \cdots n_k$.

Worked example: outfits as a decision tree

Question. There are three shirts and two pairs of pants. How many outfits can be made?

Solution. Every shirt can be paired with either pair of pants, so there are $3 \cdot 2 = 6$ outfits. The tree displays all outcomes.



Worked example: license plates with repeated characters

Question. How many strings have two uppercase letters followed by three digits, allowing repetition?

Solution. The choices for the five positions are $26, 26, 10, 10, 10$, giving $26^2 \cdot 10^3 = 676000$. Each position is an independent choice.

3 The addition principle

If one outcome belongs to exactly one of several disjoint categories, add the numbers of outcomes in the categories. If the categories overlap, ordinary addition double-counts the overlap.

Worked example: choosing a single elective

Question. A student chooses one course from 8 mathematics courses or 5 philosophy courses. How many choices exist, assuming the lists do not overlap?

Solution. The categories are disjoint, so $8 + 5 = 13$ choices. This is addition because the student chooses *one from either list*, not one from each.

Worked example: overlapping choices

Question. Of 20 students, 12 study Java, 9 study Python, and 5 study both. How many study at least one?

Solution. Let J and P denote the two groups. By inclusion–exclusion, $|J \cup P| = 12 + 9 - 5 = 16$. Subtract the five counted twice.

4 The complement principle

Sometimes counting the forbidden outcomes is easier than counting the desired ones directly. If U is the set of all possible outcomes and B is the set of bad outcomes, then $|U \setminus B| = |U| - |B|$.

Worked example: at least one digit is zero

Question. How many four-digit strings over $\{0, \ldots, 9\}$ contain at least one zero? Leading zeros are allowed.

Solution. There are 10^4 total strings. Exactly 9^4 contain no zero, since every position then has nine choices. The answer is $10^4 - 9^4 = 3439$.

5 Factorials and arranging distinct objects

For $n \geq 1$, $n! = n(n-1)\cdots 2 \cdot 1$ and $0! = 1$. To arrange n distinct objects in a row, the first position has n choices, the next $n - 1$, and so on. Therefore the number of arrangements is $n!$.

Worked example: arrange five books

Question. In how many orders can five distinct books be placed on a shelf?

Solution. There are $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5! = 120$ orders.

6 Permutations: choosing and ordering

An r-permutation of n distinct objects is an *ordered* selection of r objects without replacement. For $0 \leq r \leq n$,

$$P(n, r) = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}.$$

Worked example: assigning officer positions

Question. From 8 students, choose a president, vice president, and treasurer. How many assignments are possible?

Solution. The positions have different responsibilities, so order matters. There are $8 \cdot 7 \cdot 6 = 336$ assignments.

7 Combinations: choosing without ordering

An r-combination is a subset of size r chosen from n distinct objects. Its number is

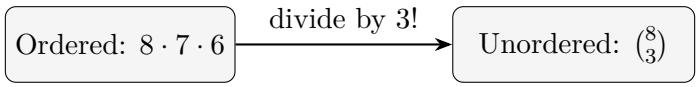
$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

The $r!$ in the denominator removes the $r!$ orders of each selected group.

Worked example: forming a committee

Question. How many three-person committees can be formed from eight students?

Solution. The order in which committee members are named does not matter, so $\binom{8}{3} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56$.



8 Proving the symmetry identity

Question. Prove $\binom{n}{r} = \binom{n}{n-r}$ for $0 \le r \le n$.

Proof. Let S be a set of n distinct elements. Every subset $A \subseteq S$ of size r determines a unique complement $S \setminus A$ of size $n - r$. Conversely, every $(n-r)$-element subset is the complement of exactly one r-element subset. Thus the two collections have the same size, so $\binom{n}{r} = \binom{n}{n-r}$. ■

Why the proof works. The proof constructs a bijection: each chosen group corresponds to the group left out.

9 Counting with restrictions

Worked example: arrangements with two adjacent people

Question. Six distinct people stand in a line. In how many orders are Alice and Bob adjacent?

Solution. Treat Alice and Bob as one block. There are now five objects to arrange, giving 5! orders. Within their block, Alice and Bob can appear in either order. Thus $2 \cdot 5! = 240$.

Worked example: committees with a required member

Question. How many four-person committees from ten students contain Alice?

Solution. Include Alice, then choose the other three from the remaining nine: $\binom{9}{3} = 84$.

10 Practice questions and complete solutions

Question. 1. A password has three letters followed by two digits, with no repeated letters or digits. How many passwords exist?

Solution. The choices are $26, 25, 24, 10, 9$, so the total is $26 \cdot 25 \cdot 24 \cdot 10 \cdot 9 = 1,404,000$. **Question.** 2. How many subsets of a ten-element set have exactly two elements?

Solution. Choose two elements without ordering: $\binom{10}{2} = 45$. **Question.** 3. How many six-bit strings contain at least one 1?

Solution. Of the $2^6 = 64$ bit strings, exactly one has no 1 (the all-zero string). Therefore $64 - 1 = 63$. **Question.** 4. Prove $\binom{n}{0} = 1$.

Proof. For any set S of size n, there is exactly one subset with zero elements, namely $\varnothing$. Hence the number of ways to choose zero elements is one. ■

11 Common mistakes

Do not multiply when choosing *either* of two disjoint categories. Do not add when choosing one item from *each* category. Do not use permutations for committees, and do not use combinations when roles differ. For complements, count the entire sample space consistently, including whether leading zeros are allowed.